

Theorem 2-1

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A given connected graph G is an Euler graph iff all vertices of G are of even degree

Proof (necessary part)

Suppose that G is an Euler graph.
i, G contains a closed walk which contain every edge of G exactly once.
(Euler line)

In tracing this walk we observe that every time the walk meets a vertex ' v ' it goes through two new edges incident on v - with one we 'entered' v and the other 'exited'.

This is true not only for all intermediate vertices of the walk but also of the terminal vertex, because we 'exited' and 'entered' the same vertex at the beginning and end of the walk respectively.

Thus if G is an Euler graph, the degree of every vertex is even. Now, sufficient part,

Assume that all vertices of G are even degree. Now we construct a walk starting at an arbitrary vertex v and going through the edges of G such that no edge is traced more than once.

Since every vertex is of even degree we can enter from every vertex we enter, and stop the tracing at the vertex v . Let this closed walk be w . If w includes all edges of G , then G is an Euler graph.

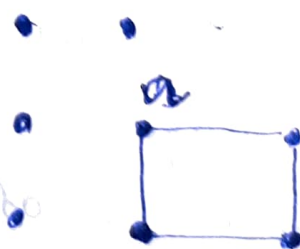
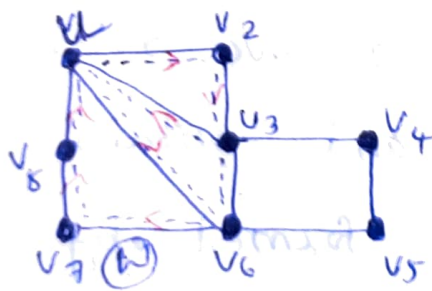
If not we remove from G , all the edges in w and obtain a subgraph H of G formed by the remaining edges.

Since both G & w have their vertices of even degree, degree of

vertices of H are also even. Moreover H must touch W at least at one vertex 'a' because G is connected. Starting from 'a' we can construct a new walk in H terminating at 'a' and this walk in H can be combined with W to form a new walk which starts and ends at vertex v and has more edges than W .

This process can be repeated until we obtain a closed walk that travels through all edges of G exactly once.

Thus G is an Euler graph.



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